

18.06 Pset 1

Due Friday Sep 16 at 11am on Gradescope. Submit in PDF format: a decent-quality **scan/image** of any handwritten solutions (e.g. get a scanner app on your phone or use a tablet), and a PDF printout of your Jupyter notebook showing your code and (clearly labeled) results.

Problem 1 (15 points)

Suppose A is a 3×3 matrix, B is a 2×3 matrix, x is a 3-component column vector, and y is a 2-component column vector.

What is the **shape** of the output (e.g. scalar, 4×3 matrix, 5-component column vector, etcetera) of the following operations, or say **nonsense** if the operation doesn't make sense.

1. AB
2. BA
3. ABx
4. Bx
5. $y^T Ax$
6. $y^T Bx$
7. $x^T y$
8. yx^T
9. yx
10. $A^2 = AA$
11. $B^2 = BB$
12. $\frac{x}{A}$
13. $\frac{x}{x}$
14. $\frac{x}{3}$
15. $\frac{B}{3}$

Problem 2 (3+3+3+3+3 points)

Given a function $f(x)$, we can *sample* it at n equally spaced points

$x_1 = a, x_2 = a + \Delta x, x_3 = a + 2\Delta x, \dots, x_n = a + (n-1)\Delta x = b$ from a to b to obtain a column vector

$$\vec{f} = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{pmatrix},$$

where $f_k = f(x_k) = f(a + (k-1)\Delta x)$.

(a) An *approximate derivative* of $f(x)$ (called a [finite-difference approximation](#)) is given by the *difference* operation:

$$D\vec{f} = \frac{1}{\Delta x} \begin{pmatrix} f_2 - f_1 \\ f_3 - f_2 \\ \vdots \\ f_n - f_{n-1} \end{pmatrix}$$

Write down this linear operator D as a matrix (what size?) for $n = 5$.

(b) Construct the *same* D matrix in Julia, for $\Delta x = 1$, using the `diagm` [function](#) to build matrices out of diagonal entries, and the `fill` [function](#) to make vectors filled with constants. For example, this is a 6×8 matrix with 1's, 4's, and 7's on three diagonals:

```
In [ ]: using LinearAlgebra
        diagm(6,8, 0 => fill(1,6), 1 => fill(4,6), -2 => fill(7,4))
```

```
In [ ]: D = ??? # FILL IN YOUR CODE HERE
```

(c) Use your same code to construct a D matrix for $n = 100$ and apply it to approximately differentiate $f(x) = \sin(x)$ for $a, b = 0, 5$, and plot your result using the code below. It should look like a cosine!

```
In [ ]: a, b = 0, 5
n = 100
Δx = (b - a) / (n - 1) # spacing for n points from a to b

x = range(a, b, length=n) # array of n points from a to b
f = sin.(x) # apply sin(x) elementwise to the vector x

D = ??? # FILL IN YOUR CODE HERE

# plot f and D*f
using PyPlot
plot(x, f, "r-")
plot(x[1:end-1] .+ Δx/2, D*f, "b-")

title(L"$f(x) = \sin(x)$ and approximate derivative")
xlabel(L"x")
legend([L"f(x)", L"df/dx"])
```

(d) An *approximate integral* of a function $g(x)$ sampled at m points with spacing Δx is given by a *sum*:

$$S\vec{g} = S \underbrace{\begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_m \end{pmatrix}}_g = (g_1 + g_2 + \cdots + g_m)\Delta x,$$

where $g_k = g(x_k)$ similar to $\vec{f}$. Write the linear operator S as a matrix for $m = 6$.

(e) What is the product SD for $n = 6$ and $m = \rule{1cm}{0.4pt}$ (for the same Δx)? (That is, the approximate "integral of the derivative" is what linear operator?)

Problem 3 (5+5 points)

B is a given $n \times n$ matrix, y is a given n -component vector, and x is an unknown n -component vector satisfying the equations:

$$B(x - y) = \frac{x + y}{2}$$

(a) Give a matrix A (in terms of B and/or the $n \times n$ identity matrix I) and a vector b (in terms of y and/or B) such that x is a solution to $Ax = b$. (That is, we want to rewrite our equations in "standard" form so that we can exploit existing algorithms. Re-arrange the equation above so that all the terms involving x are on one side and all of the known vectors are on the other side.)

(b) In Julia, $Ax = b$ can be solved with `x = A \ b`. Check that your A and b from part (a) are correct (satisfy the original equation) on a random 5×5 problem. (Note: Julia has a built-in constant `I` that denotes a context-dependent identity matrix, e.g. `A + 2I` in Julia computes $A + 2I$.)

```
In [ ]: B = randn(5,5) # a random 5x5 matrix
        y = randn(5)   # a random 5-component vector

        A = ??? # FILL IN YOUR CODE HERE from your answer in (a)
        b = ??? # FILL IN YOUR CODE HERE from your answer in (a)

        x = A \ b # solve Ax=b

        # check: does B(x - y) equal (x+y)/2 , up to roundoff errors?

        B*(x - y) ≈ (x + y)/2 # ≈ is equality up to roundoff ... should return "true"
```

Problem 4 (5+5 points)

(From Strang, section 2.2, problem 14.) Consider Gaussian elimination on the following system of equations:

$$2x + 5y + z = 0 \tag{1}$$

$$4x + dy + z = 2 \tag{2}$$

$$y - z = 3 \tag{3}$$

(Write your solution in matrix form.)

- What number d forces you to do a row exchange during elimination, and what (non-singular) triangular system do you obtain for that d ?
- What value of d would make this system singular (no third pivot, i.e. no way to get a triangular system with 3 nonzero values on the diagonal)?

Problem 5 (10 points)

(From Strang, section 2.2, problem 11.)

A system of linear equations $Ax=b$ cannot have *exactly two* solutions. An easy way to see why: if two vectors x and $y \neq x$ are two solutions (i.e. $Ax=b$ and $Ay=b$), what is another solution? (Hint: $x+y$ is almost right.)

Problem 6 (5+5+5 points)

Suppose we want to solve $Ax = b$ for **more than one right-hand side** b . For example, suppose

$$A = \begin{pmatrix} 1 & 6 & -3 \\ -2 & 3 & 4 \\ 1 & 0 & -2 \end{pmatrix}$$

and want to solve *both* $Ax_1 = b_1$ and $Ax_2 = b_2$ for the right-hand sides:

$$b_1 = \begin{pmatrix} 7 \\ 3 \\ 0 \end{pmatrix} \quad b_2 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

(a)

Explain why solving *both* $Ax_1 = b_1$ and $Ax_2 = b_2$ is equivalent to solving $AX = B$ where X is an unknown *matrix* (of what shape?) and B is a given matrix on the right-hand-side. Give B explicitly, and relate X to your desired solutions x_1 and x_2 .

(Hint: how does matrix multiplication AX relate to multiplying A by each column of X ?)

(b)

Solve your $AX = B$ equation by forming the augmented matrix $(A \ B)$, reducing it to upper-triangular form (once), and doing backsubstitution (twice) to obtain X and hence x_1 and x_2 .

(c)

You can solve $AX = B$ in Julia by the code `X = A \ B`. The matrix A is given below in Julia. Enter the matrix B , compute `X = A \ B`, and verify that it matches the answer you computed by hand in (b).

```
In [ ]: A = [ 1  6 -3
             -2  3  4
              1  0 -2 ]
```

```
In [ ]: B = [ ???  
            ???  
            ??? ]
```

```
In [ ]: X = A \ B    # solve  $AX = B$  for  $X$ 
```